

PSOC™ Edge E84: Extended Boot training manual

About this document

Scope and purpose

This document is a training manual for the hands-on exercise with an Extended Boot on PSOC™ Edge using ModusToolbox™.

Intended audience

The document is intended for design engineers, technicians, and developers of electronic systems.

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Introduction

1 Introduction

This manual provides specific instructions to create, configure, build, and run the code examples on the PSOC™ Edge E84. This lab session will cover:

- How to replace the Extended Boot on the device with a new version
- Extended Boot capability to copy an image from the external flash to SRAM and launch it from SRAM

Required development tools

2 Required development tools

The hands-on training requires either KIT_PSE84_EVAL_EPC2 or KIT_PSE84_AI.

- ModusToolbox™ software v3.8 or later
 - Recommended installation via [ModusToolbox™ Setup tool](#)
- Eclipse IDE for ModusToolbox™ v2026.3.0 or later
 - Recommended installation via [ModusToolbox™ Setup tool](#)
- Edge Protect Security Suite v1.6.1 or later.
 - Installed by [ModusToolbox™ Setup tool](#) as a dependency of ModusToolbox™
- ModusToolbox™ Programming Tools v1.8.1 or later
 - Installed by [ModusToolbox™ Setup tool](#) as a dependency of ModusToolbox™
- Board support package (BSP):
 - KIT_PSE84_EVAL_EPC2: v1.3.0 or later (available with ModusToolbox™)
 - KIT_PSE84_AI: v1.3.0 or later (available with ModusToolbox™)
- Terminal emulator
 - Instructions in this document use Tera Term
- [PSOC™ Edge E84 Evaluation Kit](#) (KIT_PSE84_EVAL)
 - Ensure device boots from external memory: +
 - BOOT SW/SW6 in 'ON/HIGH' position
 - Disable alternate serial interface configuration:
 - J20/J21 in Tristate/Not-Connected (NC) position
- [PSOC™ Edge AI Evaluation Kit](#) (KIT_PSE84_AI)

Prerequisites

3 Prerequisites

Install all required software and hardware listed in the [Required development tools](#) section.

Note that this class will not cover the basics of ModusToolbox™ and PSOC™ Edge.

- For an introduction to PSOC™, including a getting started guide to ModusToolbox™, go to: <https://www.infineon.com/product-information/psocdeveloper>
- For PSOC™ Edge trainings, from beginner tutorials to advanced trainings, go to: [PSOC™ Edge E84 Training Collection](#)

Replacing Extended Boot

4 Replacing Extended Boot

4.1 Pre-requisite

To replace the Extended Boot, ownership of the device should already be established via the provisioning process. Training Manual for getting started with PSOC™ Edge E84 Security using ModusToolbox™ provides detailed step-by-step instructions. If those steps have not yet been completed, they should be finished before proceeding further with this lab session.

4.2 Objective

The objective of this lab session is to understand how to replace the Extended Boot on PSOC™ Edge devices. The Extended Boot can be replaced on the device with the version that is shipped with Edge Protect tools.

4.3 Detailed solution

1. Follow the step-by-step instructions provided in the Training Manual for getting started with PSOC™ Edge E84 Security using the ModusToolbox™ guide to obtain the device ownership. If the steps to obtain ownership have already been completed, re-obtaining ownership is not required
2. Run the command below to observe the version of the Extended Boot

```
edgeprotecttools -t pse84 device-info
```

3. Open the policy file *policy_oem_provisioning.json* located in *<application-directory>/policy/* directory
4. Update the *extended_boot_image* policy with an absolute path to the Extended Boot image. For example, see the “value” field in the policy below, which points to the new version of the Extended Boot image

```
"extended_boot_image": {  
    "description": "Image of Extended Boot in CBOR format, signed by IFX",  
  
    "value": "C:/WS/PSOC_Edge_Basic_Secure_App/packets/apps/extended_boot/extended-boot-serial.cbor"  
}
```

5. Now, provision the device using the following command and observe the log message that indicates provisioning completed successfully

```
edgeprotecttools -t pse8xs2 provision-device -p  
policy/policy_oem_provisioning.json --key keys/oem_private_key_0.pem --ifx-  
oem-cert packets/apps/prov_oem/oem_cert.bin
```

Replacing Extended Boot

```
: O : INFO : ** Detected Device: PSE846GPS2DBZC4A
: O : INFO : ** SRAM Boot version: 2.0.0.6022
: O : INFO : ** RRAM Boot version: 2.0.0.7127
: O : INFO : ** SE RT Services Base version: 1.0.0.2361
: O : INFO : ** SE RT Services version: 0.0.0.0
: O : INFO : ** Extended Boot version: 1.1.0.1700
: O : INFO : ** Boot Status : CYBOOT_SUCCESS
: O : INFO : ** Life Cycle : DEVELOPMENT
: O : INFO : Info : [catld.sys] Examination succeed
: O : INFO : Info : gdb port disabled
: O : INFO :
: O : INFO :      TargetName          Type           Endian TapName          State
: O : INFO :      -----
: O : INFO :      --
: O : INFO :      0* catld.sys          mem_ap        little catld.cpu        halted
: O : INFO : Info : Listening on port 6668 for tcl connections
: E : INFO : Chip lifecycle stage: DEVELOPMENT
: E : INFO : ROM Boot is ready for DLM package upload
: E : INFO : Load application 'C:\workspace\Security_Intro\Basic_secure_app\packets\apps\prov_oem\cyapp_prov_oem_dlm.he
x'
: E : INFO : Validation of the DLM package success
: E : INFO : Start DLM package execution
: E : INFO : Application execution completed with status code: (0xf2a00001) - CYAPP_SUCCESS: The provisioning applicati
on completed successfully
: E : INFO : Chip lifecycle stage: DEVELOPMENT
: E : INFO : *****
: E : INFO : PROVISIONING PASSED
: E : INFO : *****
```

Figure 1 **Terminal output for provisioning**

- On the successful completion of the provisioning, the Extended Boot is replaced. The updated version of the Extended Boot will be displayed on the console, as shown in [Figure 2](#). Compare this version with the one shown before the replacement to confirm the update

```
: INFO : ** Silicon: 0xED94, Family: 0x115, Rev.: 0x21 (B0)
: INFO : ** Detected Device: PSE846GPS2DBZC4A
: INFO : ** SROM Boot version: 2.0.0.6022
: INFO : ** RRAM Boot version: 2.0.0.7127
: INFO : ** SE RT Services Base version: 1.0.0.2361
: INFO : ** SE RT Services version: 0.0.0.0
: INFO : ** Extended Boot version: 1.1.0.1700
: INFO : ** Boot Status : CYBOOT_SUCCESS
: INFO : ** Life Cycle : DEVELOPMENT
```

Figure 2 **Extended Boot version**

Note: Replacement of Extended Boot is not necessarily an update or downgrade of Extended Boot. The image can be replaced on the device with the same version.

4.3.1 Conclusion

The Extended Boot on the device has been replaced with the latest version provided in the Edge Protect tools.

SRAM loading of the user application using the Extended Boot

5 SRAM loading of the user application using the Extended Boot

5.1 Prerequisite

The ownership transfer should have been transferred via the provisioning process in the previous lab, Training manual for getting started with PSOC™ Edge E84 Security using ModusToolbox™.

5.2 Objective

By default, the 'Basic Secure Application' code example is configured to program and run from external flash. All three projects (proj_cm33_s, proj_cm33_ns, and proj_cm55) in this code example are built, programmed, and executed in-place from external flash. Upon reset, the Extended Boot searches for a valid image (in MCUboot format) in the predefined location and launches it. For the Extended Boot to search for the image in external flash, the boot source must be set to external flash.

In this lab session, let us learn how to load a project from an external flash to SRAM and launch it in SRAM. Before that, let us first understand the fundamental design concepts for this use case.

5.3 Description

The 'Basic Secure Application' code example is built to boot from the external flash (XIP).

To select the boot source in external flash, the BOOT Pin and Boot Policy must be set accordingly.

1. Default policy defines oem_alt_app_address as 0x7010_0000, which is in external flash. This is the default start address of the application if external flash is selected as a boot source
2. Default policy also enables oem_alt_boot, which indicates alternate boot option is enabled
3. To select external flash as the boot source, BOOT.1 Pin (BOOT SW) (P17.6) must be in 'ON/HIGH' position

Note: This is a DIY (Do it yourself) lab exercise. See the detailed solution below only if guidance is needed.

5.4 Detailed solution

5.4.1 Creating the project

This hands-on session uses the 'Basic Secure Application' code example. This is the same code example that is used in the "Introduction to PSOC™ Edge Security" training. See the "Training Manual for getting started with PSOC™ Edge E84 Security using ModusToolbox™" for more details.

To create a new Basic Secure code example, follow the detailed instructions provided in the Introduction to PSOC™ Edge Security lab session.

Note: The code example is configured to use the GCC_ARM compiler by default. We use the same compiler in this lab session.

SRAM loading of the user application using the Extended Boot

5.4.2 Configure proj_cm33_s to run from SRAM

In the default configuration, proj_cm33_s of Basic Secure Application is built to run from an external flash. Modify the default memory map and linker to run it from SRAM.

1. Select **Basic Secure App** in the Project Explorer and open the **Device Configurator** from the Quick Panel

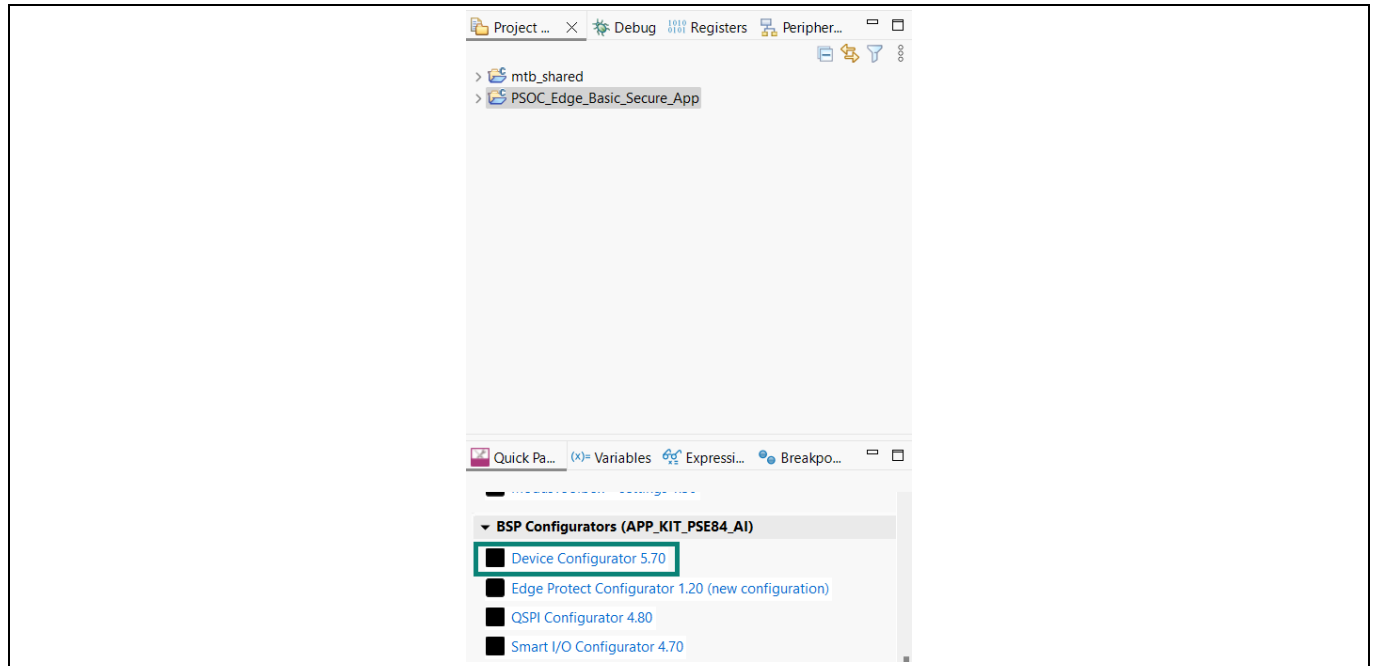


Figure 4 Open Device Configurator

2. Under the **Memory** tab, add a new region in SRAM for loading the CM33 S project

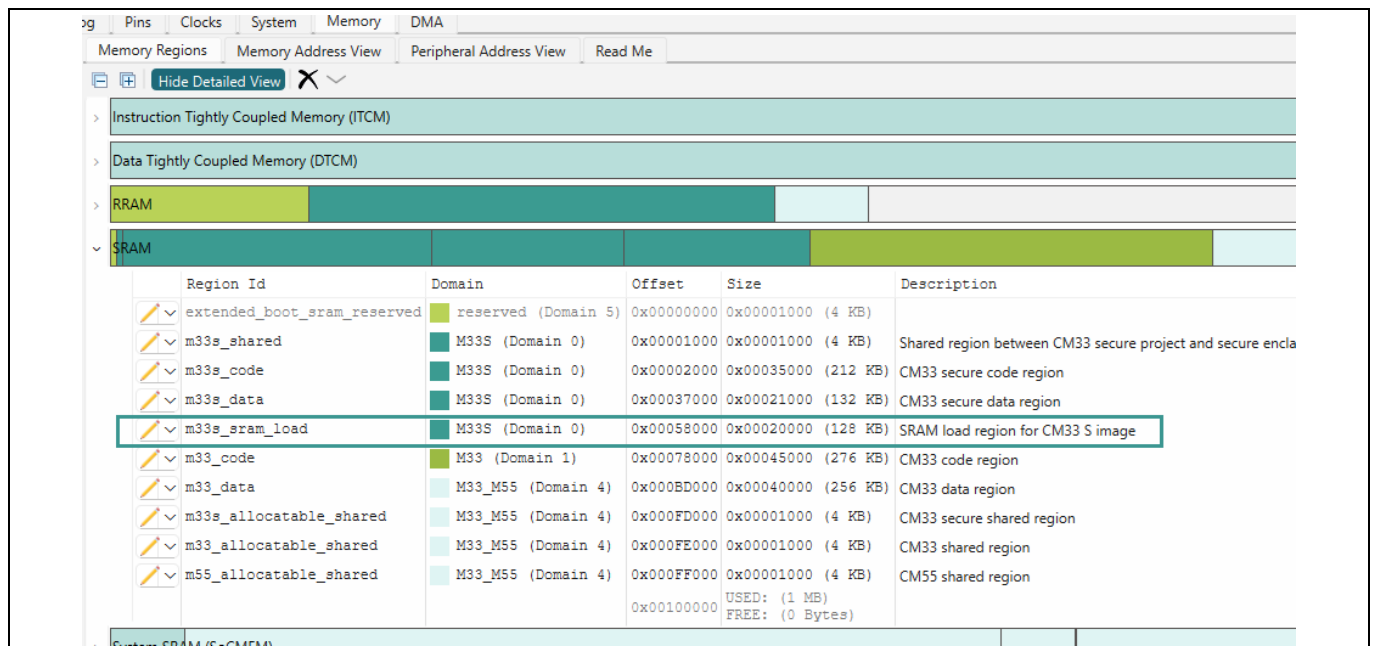


Figure 3 Create a new region for SRAM loading

Note: Adjusting the existing regions in the default memory map may be required to make space.

SRAM loading of the user application using the Extended Boot

3. Save the configuration in the **Device Configurator** and close
4. In the Basic Secure Application, open the cm33s linker file in editor
`<pathto>PSOC_Edge_Basic_Secure_App/bsps/
TARGET_APP_KIT_PSE84_EVAL_EPC2/COMPONENT_CM33/TOOLCHAIN_GCC_ARM/pse84_s_cm33.ld` and make the following change in linker file
5. Find and replace all **m33s_nvm_sel_S** with **m33s_sram_load_sel_S** and save these changes

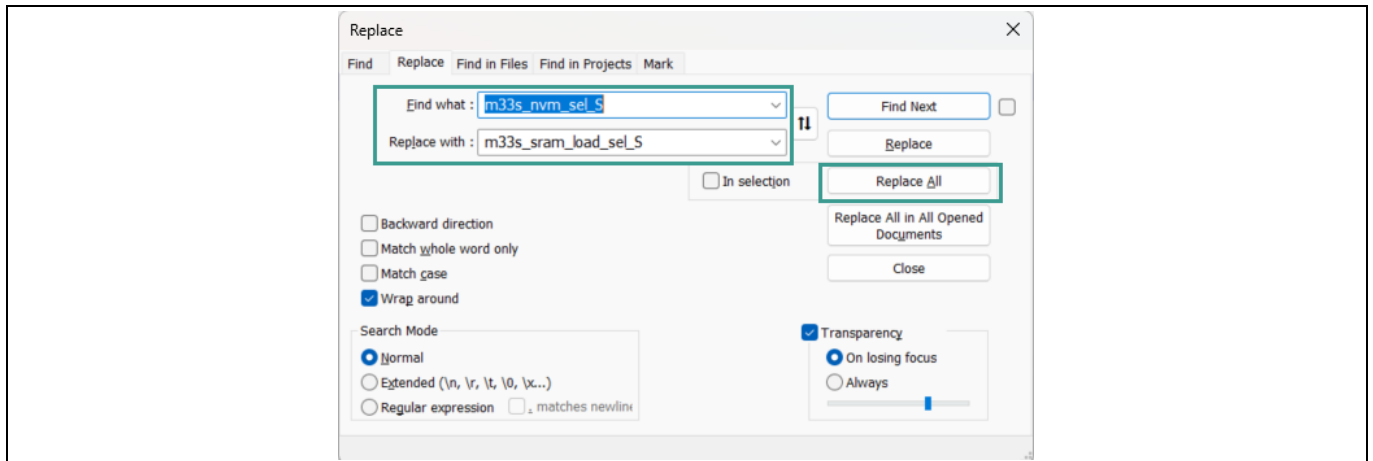


Figure 4 Edit CM33 S linker

6. Open the file `<pathto>PSOC_Edge_Basic_Secure_App/configs/boot_with_extended_boot.json` and add the following additional “load-address” input for the sign command, and save

```
{
  "command" : "sign",
  "inputs" :
  [
    {
      "file" : "../build/project_hex/proj_cm33_s.hex",
      "header-size": "0x400",
      "fill-value" : "0xff",
      "slot-size" : "0x240000",
      "hex-address" : "{{CYMEM_CM33_0_S_m33s_nvm_S_START}}",
      "load-address" : "{{CYMEM_CM33_0_S_m33s_sram_load_S_START}}"
    }
  ],
}
```

7. If the cm33_s application contains veneer files, remove them. This is required because the address of the secure project was modified after the last build

SRAM loading of the user application using the Extended Boot

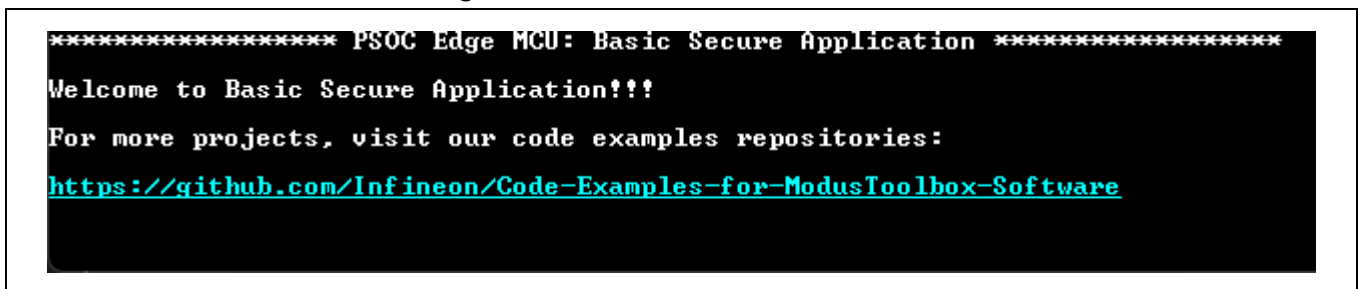
8. Build and program the Basic Secure Application in the ModusToolbox™

Note: The Instructions to build and program are the same as the previous lab in the previous session – “Introduction to PSOC™ Edge Security”. See the “Training Manual for getting started with PSOC™ Edge E84 Security using ModusToolbox™” for more details.

9. Ensure that the BOOT.1-pin (SW6) is in HIGH/ON position. This is required for the Extended Boot to determine the external flash as the boot source

5.4.3 Observing the results

1. Observe the console logs for the successful booting of the Basic Secure Application
2. Observe that the red LED is blinking on the device



```
***** PSOC Edge MCU: Basic Secure Application *****  
Welcome to Basic Secure Application!!!  
For more projects, visit our code examples repositories:  
https://github.com/Infineon/Code-Examples-for-ModusToolbox-Software
```

Figure 5 Terminal output

5.5 Conclusion

The secure project (proj_cm33_s) in the Basic Secure Application is programmed to the external flash. The Extended Boot copies it to SRAM and launches it.

Revision history

Revision history

Document revision	Date	Description of changes
**	2025-12-10	Initial release
*A	2026-03-03	Update the version of tools. Updated instructions and screenshot for lab 2.
*B	2026-05-25	Added KIT_PSE84_AI support in hands-on labs.

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Germany

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Email:

erratum@infineon.com

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